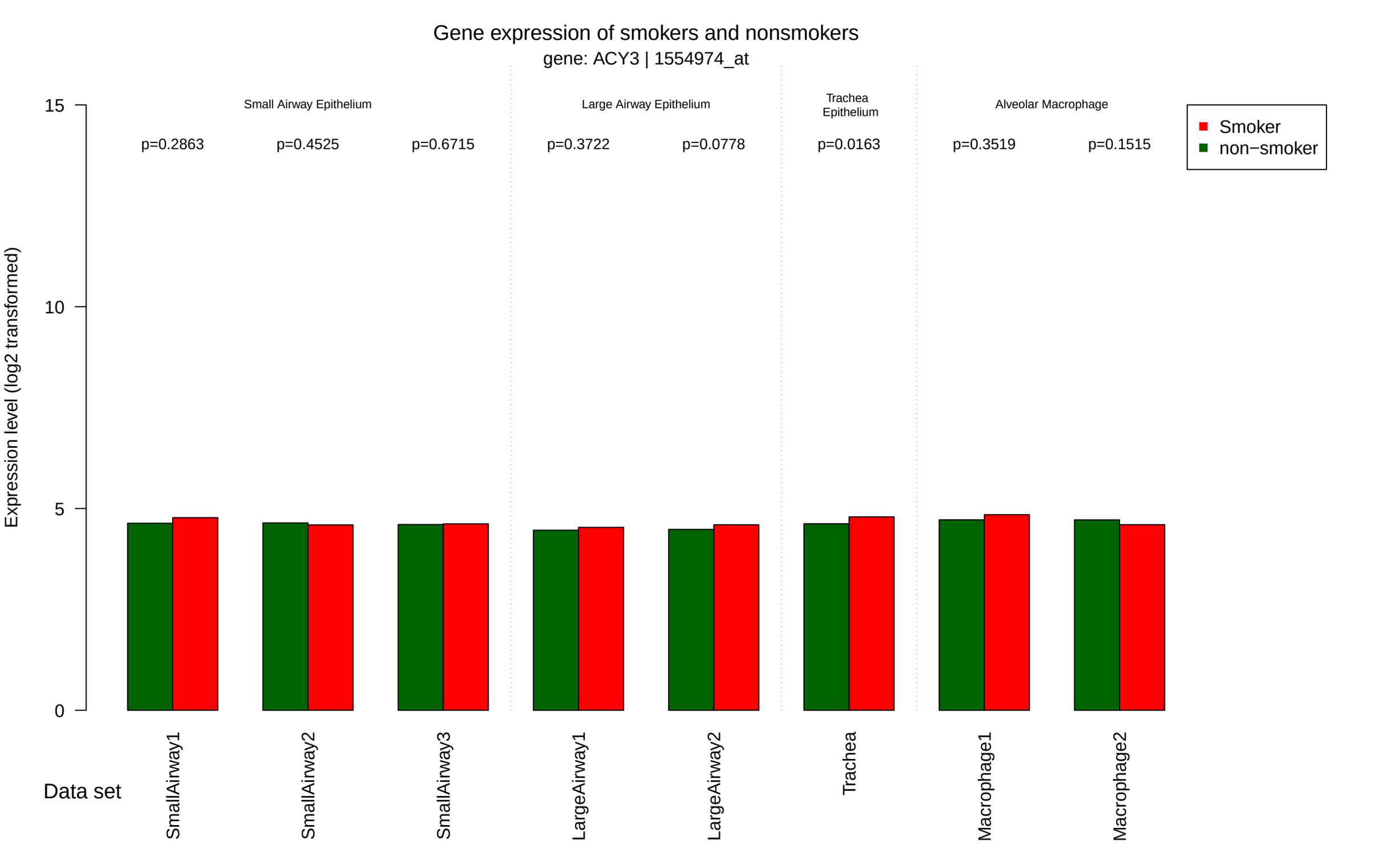
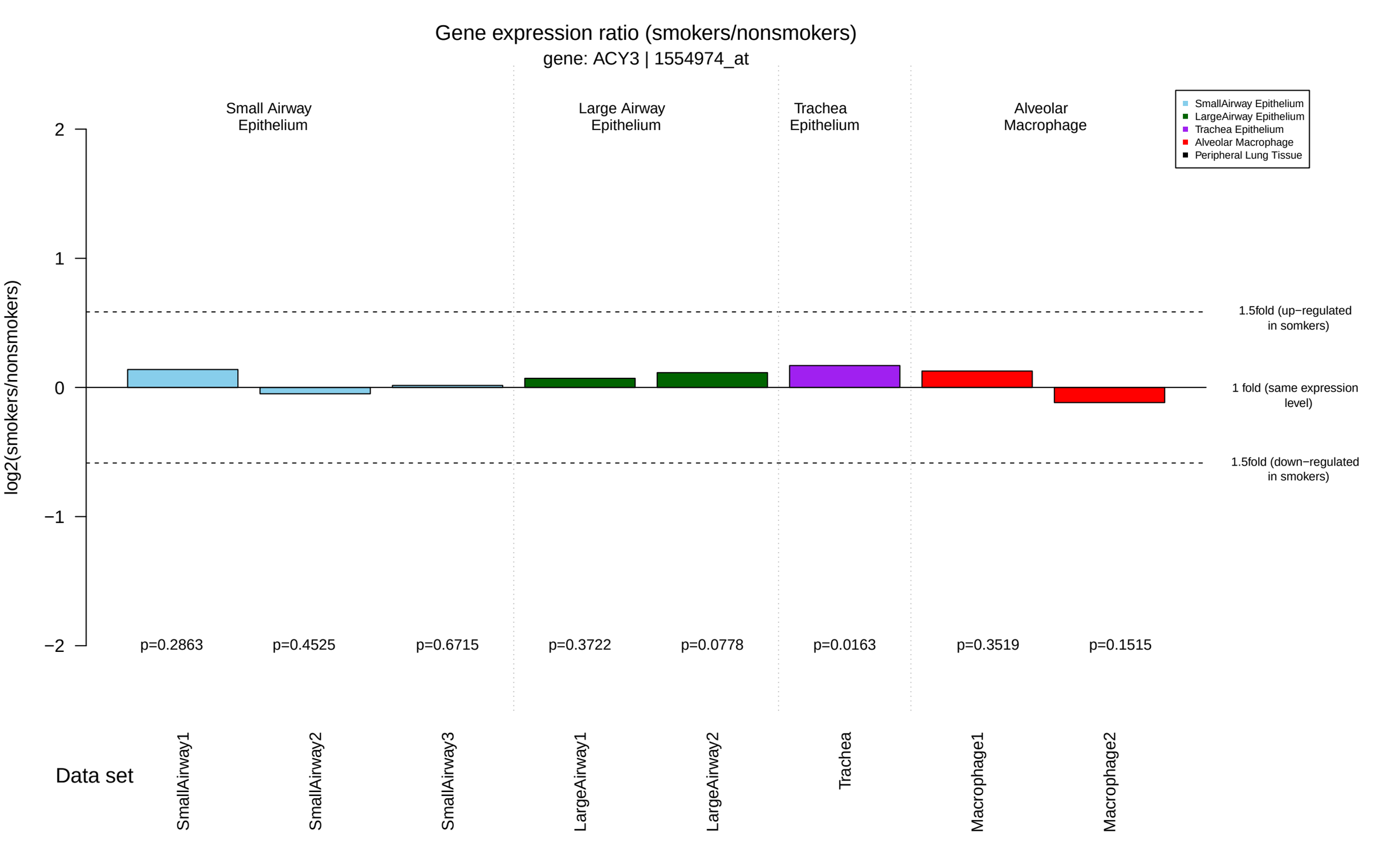
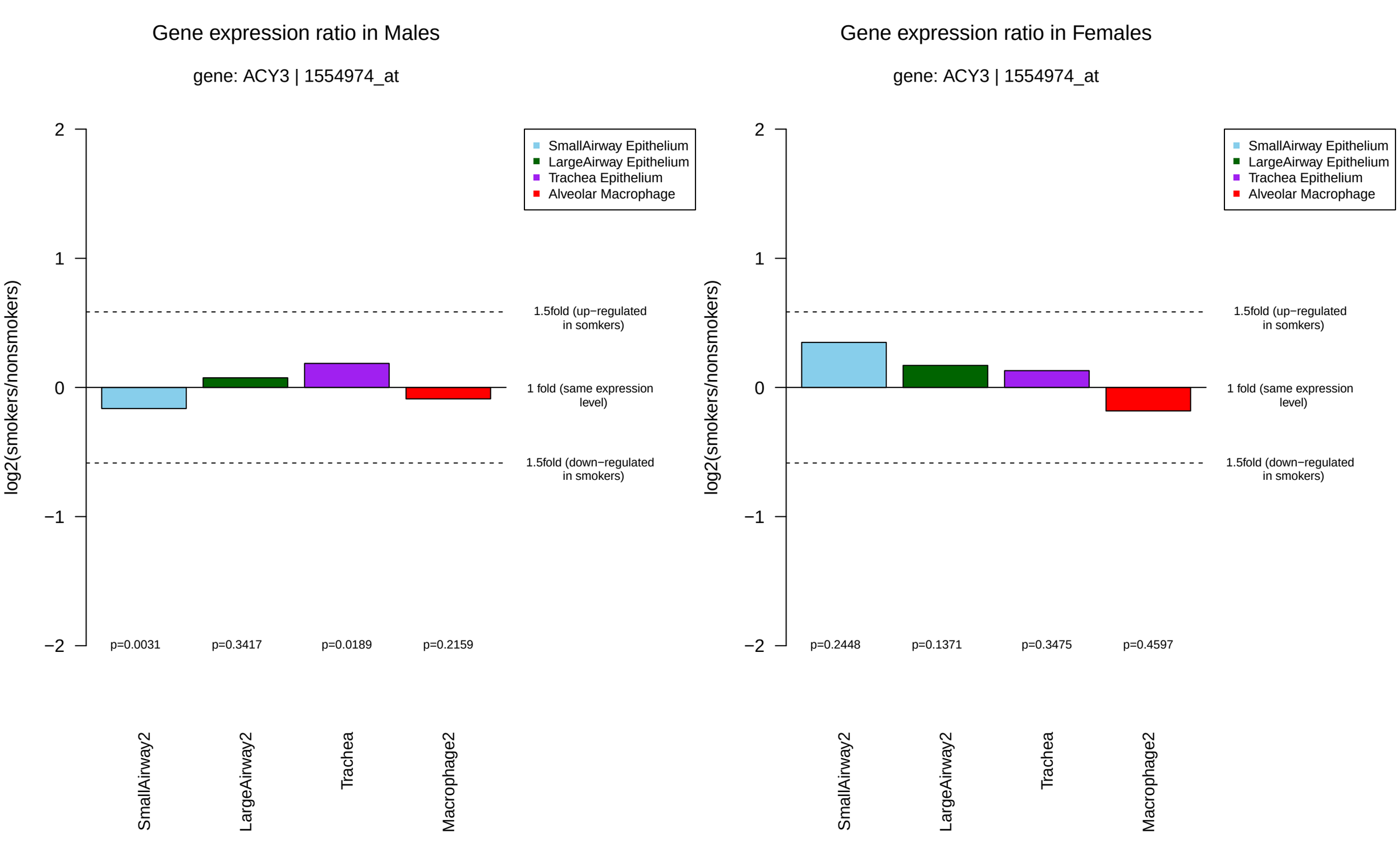
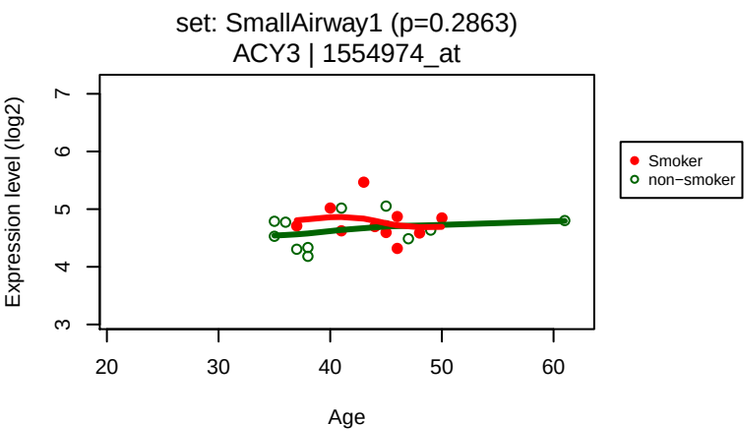
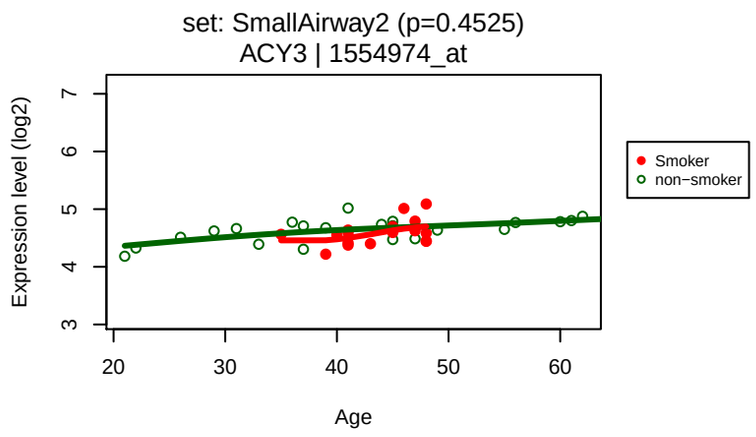
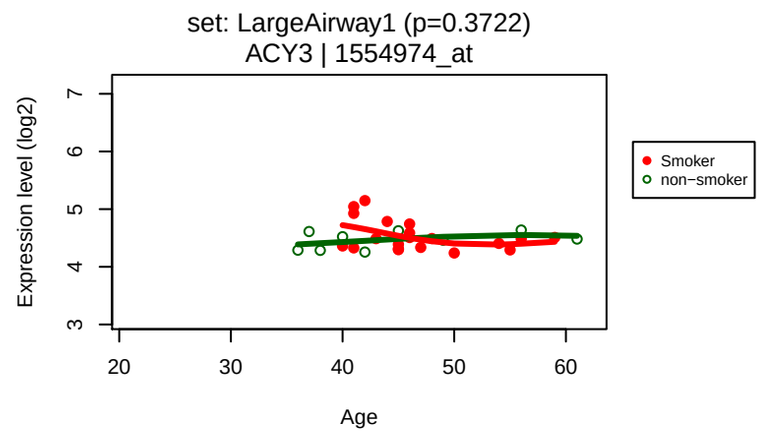

Gene expression of smokers and nonsmokers
gene: ACY3 | 1554974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2863
p=0.4525
p=0.6715
p=0.3722
p=0.0778
p=0.0163
p=0.3519
p=0.1515
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

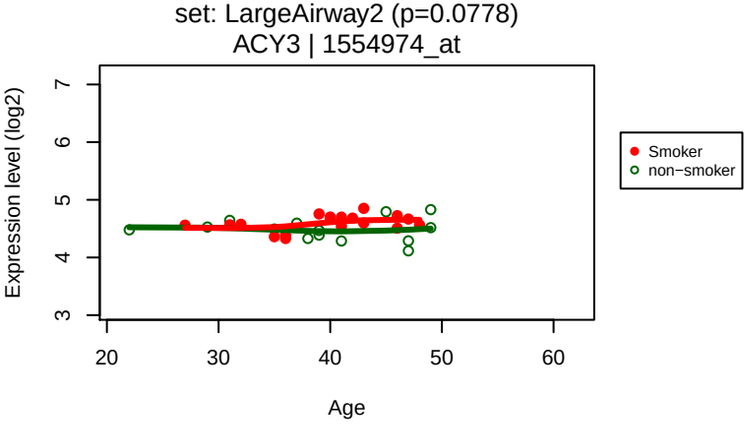

Gene expression ratio (smokers/nonsmokers)
gene: ACY3 | 1554974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2863
p=0.4525
p=0.6715
p=0.3722
p=0.0778
p=0.0163
p=0.3519
p=0.1515
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

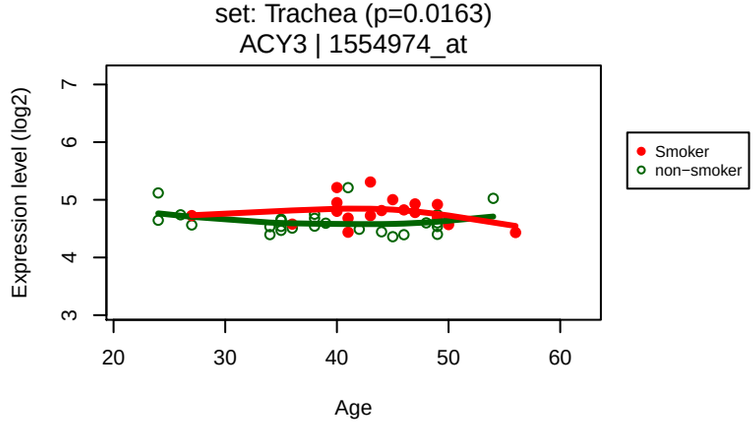

Gene expression ratio in Males
gene: ACY3 | 1554974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0031
p=0.3417
p=0.0189
p=0.2159
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACY3 | 1554974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2448
p=0.1371
p=0.3475
p=0.4597
SmallAirway2
LargeAirway2
Trachea
Macrophage2

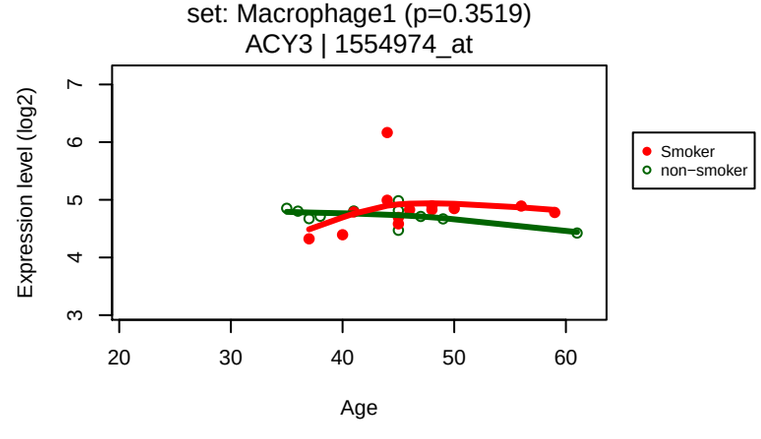

set: SmallAirway1 (p=0.2863)
ACY3 | 1554974_at
Expression level (log2)
Age
Smoker
non−smoker

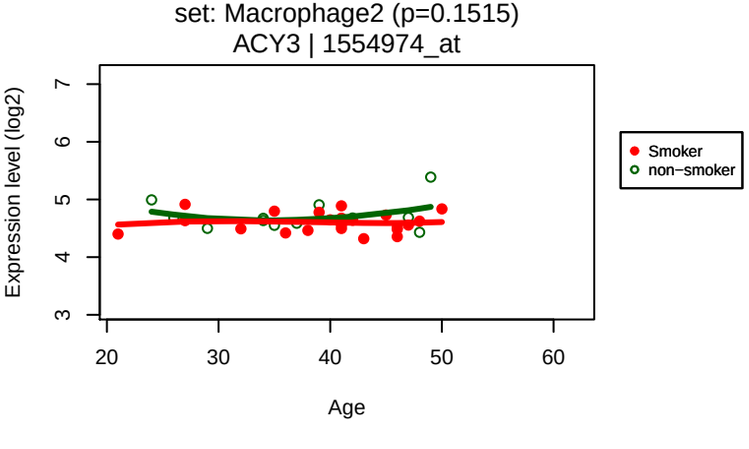

set: SmallAirway2 (p=0.4525)
ACY3 | 1554974_at
Expression level (log2)
Age
Smoker
non−smoker

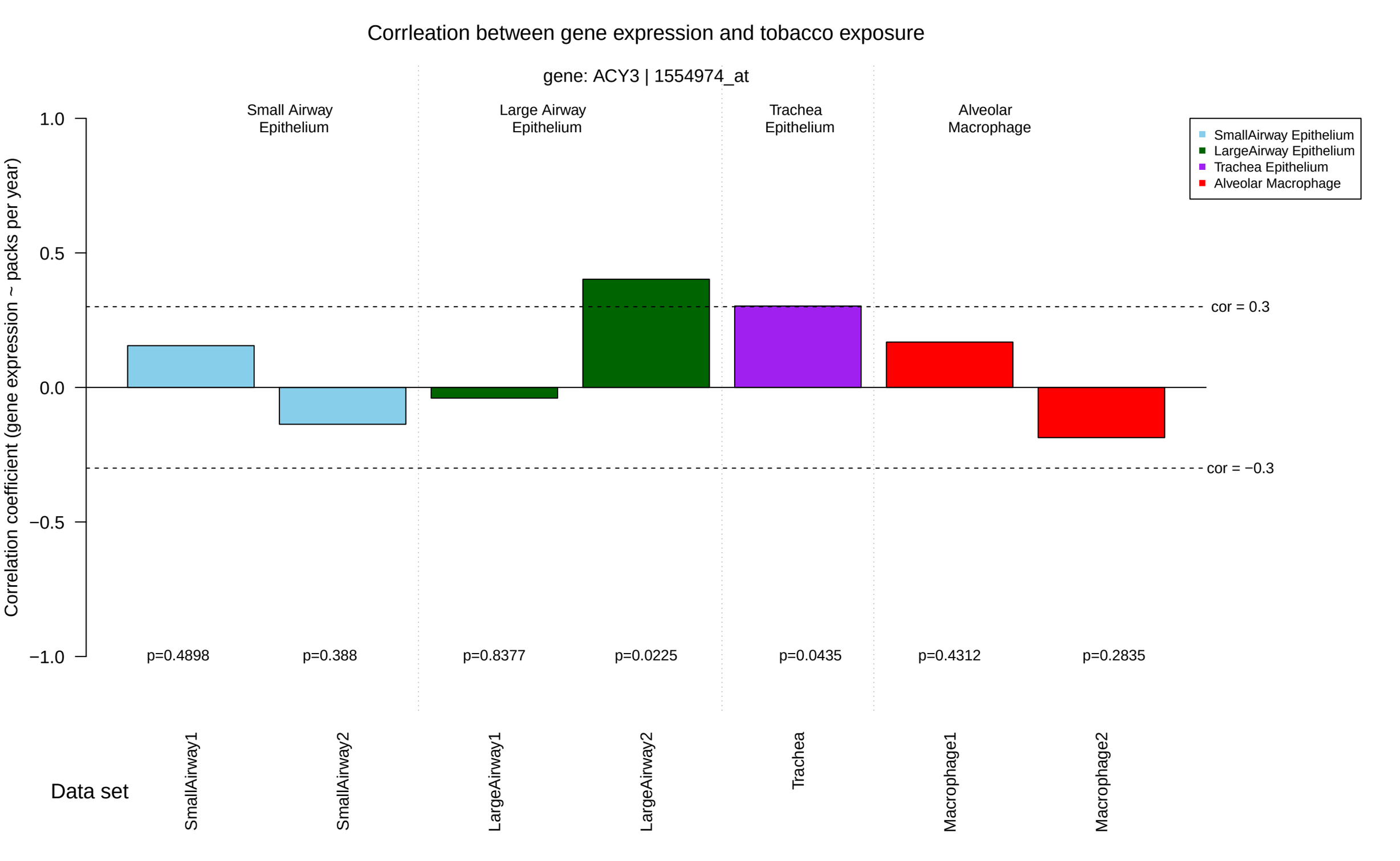

set: LargeAirway1 (p=0.3722)
ACY3 | 1554974_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0778)
ACY3 | 1554974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0163)
ACY3 | 1554974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3519)
ACY3 | 1554974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1515)
ACY3 | 1554974_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACY3 | 1554974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4898
p=0.388
p=0.8377
p=0.0225
p=0.0435
p=0.4312
p=0.2835
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2